



Ultra-compact, broadband adiabatic passage optical couplers in thin-film lithium niobate on insulator waveguides

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Abstract: We report the first demonstration of broadband adiabatic directional couplers in thin-film lithium niobate on insulator (LNOI) waveguides. A three LN-waveguide configuration with each waveguide having a ridge cross section of less than 1 square micron, built atop a layer of SiO₂ based on a 500- μ m-thick Si substrate, has been designed and constructed to optically emulate a three-state stimulated Raman adiabatic passage system, with which a unique counterintuitive adiabatic light transfer phenomenon in a high coupling efficiency of >97% (corresponding to a >15 dB splitting ratio) spanning telecom S, C, and L bands for both TE and TM polarization modes has been observed for a 2-mm long coupler length. An even broader operating bandwidth of >800 nm of the device can be found from the simulation fitting of the experimental data. The footprint of the realized LNOI adiabatic coupler has been reduced by >99% compared to its bulk counterparts. Such an ultra-compact, broadband LNOI adiabatic coupler can be further used to implement or integrate with various photonic elements, a potential building block for realizing large-scale integrated photonic (quantum) circuits in LN.

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1. Introduction

Waveguide directional couplers are one of the essential building blocks for implementing integrated photonic circuits (IPCs) for versatile applications including optical communication networking technology and the emerging photonic quantum technology [1–3]. Towards the realization of robust large-scale IPCs, components with high fabrication tolerance are in demand to increase not only the scalability but also the fidelity of the circuits. A fabrication-tolerant waveguide device usually features a low sensitivity to the operating wavelength [4,5], benefiting its application to an integrated system designed for a broadband or tunable operation. In IPCs, directional couplers can function as a beam/mode splitter, a switch, a wavelength filter, etc. [6]. However, typical directional couplers, working based on the mode interference effect occurring between two closely spaced straight waveguides via the evanescent coupling mechanism [7], are highly dependent on the operating wavelength and sensitive to fabrication errors. Several waveguide architectures including asymmetric, tapered, and bent geometries have been proposed to achieve fabrication-tolerant/broadband couplers [8,9]. These designs have been based on the engineering of a properly varied coupling coefficient along the wave propagating direction to relieve the highly dispersive mode interference condition. Beyond the scope of a mode interference coupler, a coupler scheme based on arrayed (channel) waveguides designed to

perform a counterintuitive light transfer between two well-separated outer waveguides without the possibility of direct (evanescent) coupling has been investigated to be a broadband coupling system with high fabrication tolerance [10–13]. Such a unique waveguide directional coupler works based on a peculiar adiabatic tunneling effect, which is a classical and spatial analogue of the stimulated Raman adiabatic passage (STIRAP) scheme found in quantum physics [14]. Similar to the temporal evolution of the population transition made in the STIRAP systems, an adiabatic evolution of light (in space) can be observed in such a waveguide adiabatic passage system where the light power can be transferred monotonically from one outer waveguide (input port) to the other outer waveguide (output port) with negligible excitation of the intermediate waveguide(s).

We have recently demonstrated broadband high coupling-efficiency directional couplers and broadband high extinction-ratio polarization-mode splitters in a three-waveguide and a five-waveguide adiabatic light-transfer schemes in titanium-indiffused lithium niobate (LN) waveguides, respectively [15,16]. LN has been one of the most important materials in the study and realization of the IPC technology because of its versatile applicability in building efficient passive and active functional devices due to its excellent electro-optic (EO), acousto-optic, nonlinear-optic, and piezoelectric properties and broad optical transparent region ($\sim 0.4\text{--}5\text{ }\mu\text{m}$) [17]. However, the state-of-the-art large-scale IPC technology is established based on Si photonics owing to the highly mature fabrication technologies supported by the immensely successful semiconductor industry. Because of their large refractive-index contrast, Si waveguides can be structured to dimensions that are orders of magnitude smaller than diffused waveguides in LN. Adiabatic passage waveguide systems have indeed been demonstrated in Si/CMOS photonics platforms with a miniaturized footprint of $\sim \text{mm}^2$ [18]. Though leading in the technology, Si photonics has limitations in implementing efficient and fast active elements that usually demand a quadratic nonlinearity.

A recent breakthrough evolution of the LN material platform, called thin-film LN on insulator (LNOI), has led to promising demonstrations for realizing nanoscale IPCs in this iconic quadratic nonlinear material, including wavelength converters (such as second-harmonic generators and spontaneous parametric down converters (SPDC)), EO modulators, microring resonators, and nanowire detectors [19–22]. These nanophotonic integrated circuits can be implemented because LNOI features a high-index contrast ($\Delta n \sim 0.8$ or even higher depending on the cladding material) structure (in contrast to conventional titanium-indiffused LN waveguides with $\Delta n \sim 0.01$), while retaining all the superior linear and nonlinear optical material properties. LNOI waveguide thus feature largely enhanced nonlinear conversion efficiencies [19]. In this work, we report, to the best of our knowledge, the first demonstration of a broadband adiabatic light passage process in the LNOI waveguide platform. Adiabatic directional couplers have been, e.g., proposed to be integrated with SPDC in indiffused LN waveguides to enable the simultaneous Bell state generation, pump filtering, and/or polarizing beam splitting [16,23]. Though integrated in a monolithic chip, these bulk devices have a length of several cm. In this study, we successfully realize LNOI adiabatic couplers with a footprint being reduced by $>99\%$ compared to bulk LN couplers [15]. Besides its ultra-compactness, the LNOI waveguide dispersion can be engineered with a higher flexibility via e.g., the structuring of the ridge-waveguide geometry, which is a more effective and reliable process in contrast to that through the control of the ion diffusion process in bulk LN waveguides. Benefiting from this advantage, the demonstrated LNOI couplers can work for both polarization (TE and TM) modes over a broad bandwidth, which can be of particular interest for applications in such as quantum optical systems based on polarization encoded or entangled photons.

2. Device design and simulation

For this demonstration, we study adiabatic directional couplers in a three-waveguide coupling scheme, which is an optical analogue of the simplest (three-state) STIRAP system (as illustrated in Fig. 1(a) a Λ -scheme atomic state system), using x -cut LNOI, as schematically shown in Fig. 1(b). The two outer waveguides (Wg_1 and Wg_3) are positioned parallel to the crystallographic y -axis and are well separated to inhibit a direct evanescent coupling between them, corresponding to the scenario in the STIRAP system where its first and final states ($|\varphi_1\rangle$ and $|\varphi_3\rangle$ labelled in Fig. 1(a), respectively) have the same parity (i.e., the electric-dipole transition between them is forbidden). The intermediate waveguide, Wg_2 , properly inclined with respect to the two outer waveguides, mimics the intermediate dressed state ($|\varphi_2\rangle$ labelled in Fig. 1(a)) in a STIRAP system and accomplishes the adiabatic light passage mechanism when the light is injected into Wg_1 . During the light transfer in the adiabatic coupler system shown in Fig. 1(b), the coupling strengths between Wg_1 and Wg_2 ($\kappa_{12}(y)$) and Wg_3 and Wg_2 ($\kappa_{32}(y)$) play the role of the Rabi frequencies ($\Omega_p(t)$ and $\Omega_s(t)$ labelled in Fig. 1(a)) of the pump and Stokes pulses in a STIRAP system, respectively.

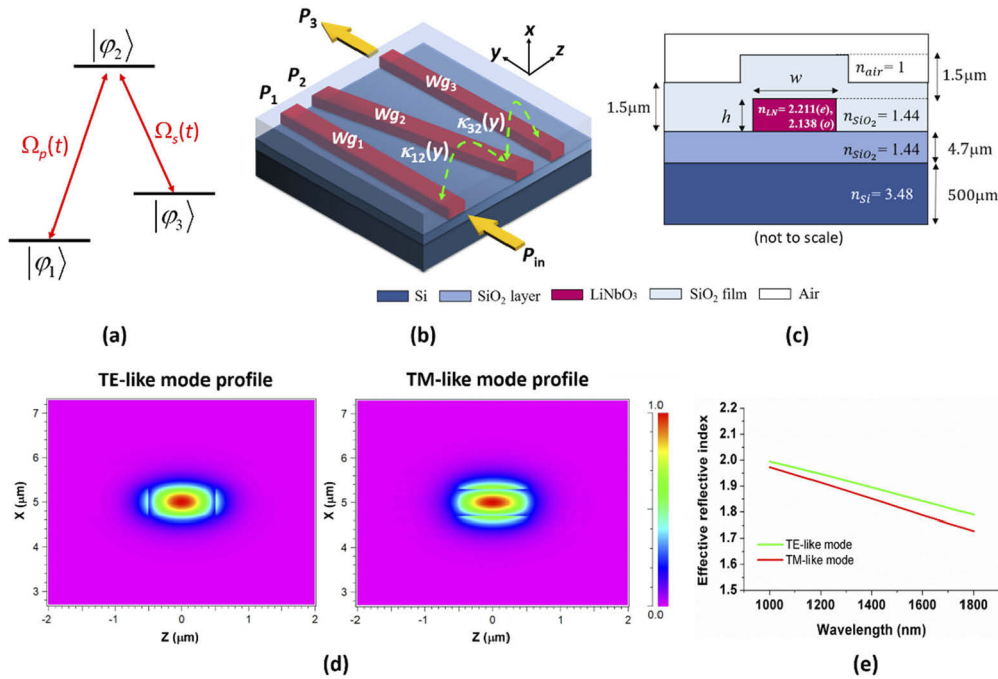


Fig. 1. (a) Schematic of a Λ -configuration STIRAP coupling system. (b) Schematic of a three-waveguide adiabatic directional coupler in x -cut LNOI. $\kappa_{12}(y)$ and $\kappa_{32}(y)$ represent the coupling strengths between Wg_1 and Wg_2 and Wg_3 and Wg_2 , respectively. (c) Schematic illustration of the cross-sectional information of the LNOI waveguide. (d) Simulated field distributions of TE-like and TM-like polarization modes in the LNOI waveguide at 1550 nm for $w = 1 \mu\text{m}$ and $h = 0.6 \mu\text{m}$. (e) Calculated dispersion curves of the LNOI waveguide with $w = 1 \mu\text{m}$ and $h = 0.6 \mu\text{m}$ for the two polarization modes.

In our design, all waveguides have a cross section described by the width w and the height h . The waveguides are patterned atop a 4.7- μm -thick SiO_2 layer and covered with a SiO_2 film of thickness 1.5 μm . The entire LNOI waveguide structure is based on a 500- μm -thick Si substrate, as illustrated in Fig. 1(c). Typical values of w range from 0.45 to 1 μm and from 0.55 to 0.6 μm for h , to support the guiding of single (fundamental) TE- and TM-like modes in the spectral

region of interest in the telecom bands from ~ 1400 to 1700 nm. Figure 1(d) shows the simulated field distributions of the two polarization modes in the LNOI waveguide at 1550 nm for $w = 1 \mu\text{m}$ and $h = 0.6 \mu\text{m}$. The result indicates that highly confined modes with a size of $\sim 0.9 \times 0.5 \mu\text{m}^2$ can be obtained in such a nanophotonic waveguide platform, which is two orders of magnitude smaller compared to a indiffused LN waveguide mode. Accordingly, the waveguide dispersion curves can be calculated, as shown in Fig. 1(e). It can be found that the effective refractive indices of the LNOI waveguide are ~ 1.856 and ~ 1.805 for TE- and TM-like modes around 1550 nm, respectively. We next investigate feasible configurations of three-waveguide coupling systems based on the information shown in Fig. 1 to design and realize compact, broadband high-efficiency adiabatic couplers in LNOI, working simultaneously for both of the polarization modes.

Figure 2(a) shows the schematic configuration of the three-waveguide LNOI coupler in the z - y Cartesian plane, where key structure parameters are labeled with l_{ac} being the length of the coupler, d_{13} , $d_{12}(y)$, and $d_{23}(y)$ being the center-to-center distances between waveguides **Wg**₁ and **Wg**₃, **Wg**₁ and **Wg**₂, and **Wg**₂ and **Wg**₃, respectively. The whole system is symmetric with respect to the center of the structure at $(z,y)=(0,0)$, i.e., $d_{12}(y)=d_{23}(-y)$, to make a complete power transfer possible in such a coupler [11]. The adiabaticity of the system is a function of the coupling strengths among the waveguides as well as the inclination of the intermediate waveguide, **Wg**₂ [11], which is determined by those structure parameters. Accordingly, we can conclude two adiabatic conditions as a criterion for building the target adiabatic coupler. One is related to the slope of the inclined **Wg**₂, $\gamma = \Delta d / \kappa_0 r l_{ac}$, which is a parameter related to the adiabatic rate of the coupling process and should be significantly smaller than unity (i.e., $\gamma < 1$; Condition A) to meet a slow adiabatic light transfer condition, where $\Delta d = d_{12}(-l_{ac}/2) - d_{23}(-l_{ac}/2)$ (or $d_{23}(l_{ac}/2) - d_{12}(l_{ac}/2)$), r is a parameter associated with the waveguide confinement (or the waveguide index contrast Δn) [24], and $\kappa_0 = \kappa_{12}(0) = \kappa_{23}(0)$ is the coupling coefficient between **Wg**₁ and **Wg**₂ or **Wg**₂ and **Wg**₃ at $y=0$ (in the center of the wave evolution in the waveguide system). The other condition, via emulating the transition scheme in a STIRAP system, gives $\kappa_{13} \sim 0$ (the coupling coefficient between **Wg**₁ and **Wg**₃) and $\kappa_{12}(-l_{ac}/2) < \kappa_{23}(-l_{ac}/2)$ (Condition B) as a guideline for defining the coupling strength among the three waveguides. A detailed analysis of these structural adiabatic conditions can be found in Refs. [11,15]. For three-waveguide adiabatic couplers built in diffused waveguides in bulk LN, a device length of several cm is found necessary to meet the adiabatic condition of $\gamma < 1$ [15] due to the relatively low $\Delta n \sim 0.01$.

We then simulate the light transfer characteristics of the three-waveguide adiabatic coupling system modeled in Fig. 2(a) using the “Beam Propagation Method” (BPM) [25]. The waveguide specifications (dimensions and refractive indices) shown in Fig. 1(c) were used to construct the waveguide systems to be investigated. Mainly, the mode field distribution and evolution in a built structure can be solved by the method in a step resolution of $0.01 \mu\text{m}$ along the transverse (z , x) directions and of $1 \mu\text{m}$ along the longitudinal (y) direction. Figures 2(b) and 2(c) show the calculated distributions of the coupling efficiency (η) of the LNOI adiabatic couplers for coupler lengths of $l_{ac}=0.4$ and 2 mm, respectively, as a function of Δd and $d_{23}(-l_{ac}/2)$ (which are two structural parameters determining the architecture of the three-waveguide coupler) for the TE-like polarized mode at 1550 nm. The distribution for the 1550 -nm TM-like mode in a $l_{ac}=2$ mm coupler is also plotted in Fig. 2(d) for comparison. Here, the coupling efficiency is defined as the ratio of the output power from the exit of **Wg**₃ to the accumulated power from the outputs of **Wg**₁, **Wg**₂, and **Wg**₃ (i.e., $\eta = P_3 / (P_1 + P_2 + P_3)$). The results clearly show that a longer device length leads to a broader distribution range of high coupling efficiency with respect to the two structural parameters Δd and $d_{23}(-l_{ac}/2)$. This implies that a high adiabaticity ($\gamma < 1$) of an adiabatic coupler can be achieved directly with a long enough device length but within an appropriate range of the inclination of **Wg**₂ and the coupling strengths among the waveguides (both are determined by the structural parameters $d_{23}(-l_{ac}/2)$ and Δd). In these calculations,

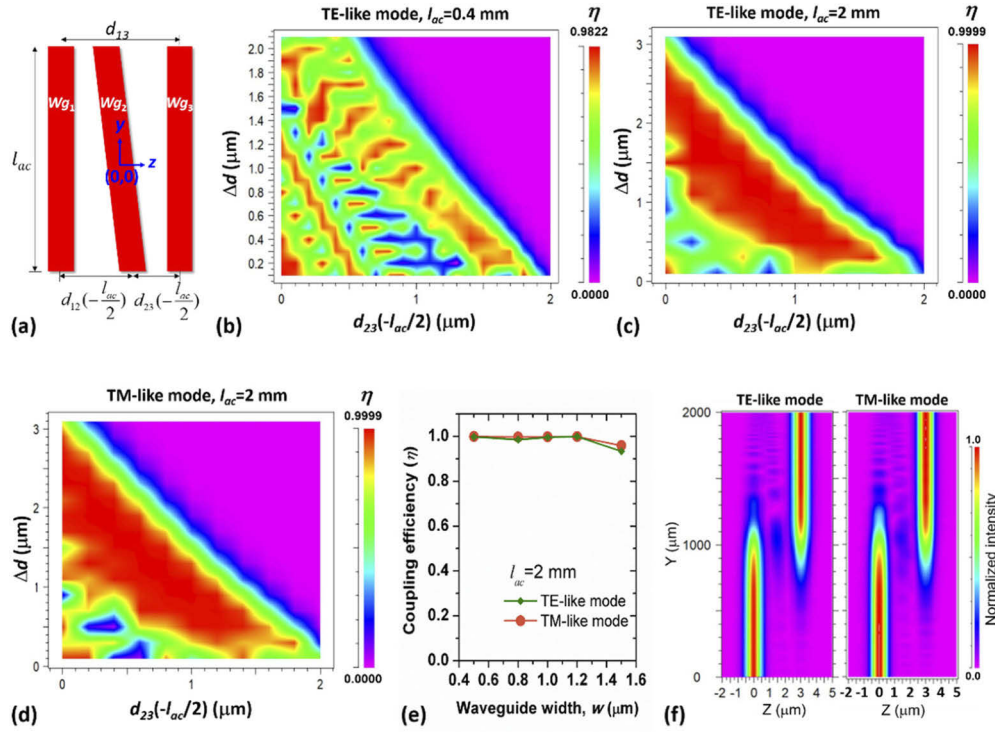


Fig. 2. (a) Schematic configuration of the three-waveguide LNOI coupler in the z - y Cartesian plane. (b) and (c) Simulated coupling efficiency (η) distributions of the LNOI adiabatic couplers as a function of $\Delta d = d_{12}(-l_{ac}/2) - d_{23}(-l_{ac}/2)$ and center-to-center distance $d_{23}(-l_{ac}/2)$ for coupler lengths of $l_{ac} = 0.4$ and 2 mm, respectively, for the TE-like polarized mode at 1550 nm. (d) Simulated coupling efficiency distribution for $l_{ac} = 2$ mm and the TM-like polarized mode at 1550 nm. (e) Calculated coupling efficiency of the 2-mm long coupler as a function of the waveguide width for TE-like and TM-like modes at 1550 nm. (f) Simulated wave evolution in a LNOI three-waveguide adiabatic coupler with $l_{ac} = 2$ mm, $w = 1$ μm , $h = 0.6$ μm , $d_{23}(-l_{ac}/2) = 1$ μm , and $\Delta d = 1$ μm for TE-like and TM-like 1550-nm modes initially excited in waveguide Wg_1 .

$w = 1$ μm and $h = 0.6$ μm were used. To further understand the influence of the waveguide width (w) on the performance of the LNOI coupler, we calculate the coupling efficiency of the 2-mm long device with respect to the variation of w for TE-like and TM-like modes at 1550 nm, as shown in Fig. 2(e). The result reveals that a high coupling efficiency of $>99\%$ (or >20 dB splitting ratio) can be maintained for the LNOI coupler with a waveguide width being in the range of from 0.5 to 1.2 μm (covering the range of interest for supporting the single fundamental mode operation) for both polarization modes. The results shown in Figs. 2(b)–2(e) manifest that a high adiabaticity coupler features a high coupling efficiency (over a broad bandwidth) and a high fabrication tolerance as aforementioned. Figure 2(f) shows the simulated wave evolution in a LNOI three-waveguide adiabatic coupler (refer to Fig. 2(a)) with $l_{ac} = 2$ mm, $w = 1$ μm , $h = 0.6$ μm , $d_{23}(-l_{ac}/2) = 1$ μm , and $\Delta d = 1$ μm for TE-like and TM-like 1550-nm modes initially excited in waveguide Wg_1 , respectively. The results clearly show an important signature of the adiabatic light transfer where most of the power has been coupled from Wg_1 to Wg_3 without an apparent excitation of Wg_2 .

3. Device fabrication and performance characterization

As shown in Figs. 2(b)–2(d), a broad range of the structure parameters ($d_{23}(-l_{ac}/2)$ and Δd) can be applied to define a three-waveguide configuration for being a high-adiabaticity counterintuitive directional coupler in LNOI especially for a longer ($>$ sub-mm) device. Interestingly, for a shorter device such as the 0.4-mm long coupler, the structure demands the use of a less inclined intermediate waveguide **Wg₂** to satisfy the adiabatic Condition **A** ($\gamma \propto \Delta d/l_{ac} \ll 1$) to achieve a high η . This can be done by reducing the Δd with the aid of employing a relatively large $d_{23}(-l_{ac}/2)$ according to the coupler scheme shown in Fig. 2(a), as also suggested/manifested by the location of the high η (red color) region on the structure-dependent efficiency distribution shown in Fig. 2(b). Δd cannot be arbitrarily small and a tradeoff should be considered between the Condition **A** and the other adiabatic condition, Condition **B**, giving $\kappa_{12}(-l_{ac}/2) < \kappa_{23}(-l_{ac}/2)$ (which prefers a smaller $d_{23}(-l_{ac}/2)$ and therefore a larger Δd). With the increase of the coupler length l_{ac} , Condition **A** can already be satisfied without relying much on the use of a reduced Δd , in this case, the tradeoff made for Δd between the two adiabatic conditions can be more relaxed, implying that a high-adiabaticity (high η) coupler can be implemented over a broad structure parameter variation range. This is revealed by the results shown in Figs. 2(b)–2(d), where the high η region of the distribution with respect to $d_{23}(-l_{ac}/2)$ and Δd gets broader with increasing l_{ac} . It is also found for the 2-mm-long coupler, that the region for $\eta > 97\%$ (region with red color) can be continuously extended to a relatively large Δd , which facilitates the access of an optimized coupler structure where both of the adiabatic conditions can be well satisfied.

We have produced several couplers with structures being in high-efficiency region of the distributions shown in Figs. 2(b)–2(d) (the following parameter ranges were accessed in this fabrication: $\Delta d = 0.2$ to $0.5 \mu\text{m}$ and $d_{23}(-l_{ac}/2) = 1.4$ to $1.7 \mu\text{m}$ for $l_{ac} = 0.4 \text{ mm}$, and $\Delta d = 0.7$ to $1.1 \mu\text{m}$ and $d_{23}(-l_{ac}/2) = 1$ to $1.2 \mu\text{m}$ for $l_{ac} = 2 \text{ mm}$). We did not consider a structure with $d_{23}(-l_{ac}/2) < 1 \mu\text{m}$ as the two waveguides, **Wg₂** and **Wg₃**, would merge at $y = -l_{ac}/2$ (i.e., forming a N-shaped waveguide system), which is challenging to produce with our current fabrication technique, though we noticed that an interesting bidirectional light transfer can be realized in such a N-shaped adiabatic waveguide structure [26]. We fabricated these LNOI waveguide structures using the ion-beam etching (IBE) method [27,28]. Mainly, we used *x*-cut LNOI substrates for this fabrication. An important advantage of using *x*-cut LN is that it is possible to be domain engineered using the electric-field poling technique at its surface [29]. The integration of a broadband adiabatic coupler with a usually narrowband (few nm or less) but tunable quasi-phase-matching device can be of interest to build novel IPC for, e.g., photonic quantum applications [16,23]. The thickness of the LN film on these LNOI substrates is 600 nm. The mask pattern layers (consisting of a tempered resist layer, a Cr layer, and a negative tone resist layer) for the IBE structuring of the three-waveguide adiabatic coupler configuration in the LNOI substrates were created by electron beam lithography and dry etching (reactive-ion etching and reactive ion beam etching in sequence). The waveguide structures of the samples were finally formed by the IBE process using an Ar ion beam of energy $\sim 400 \text{ eV}$. The etching recipe was adopted to form ridge waveguides for a height of 0.55 – $0.6 \mu\text{m}$ with relatively smooth sidewalls. The range of the waveguide height is desired to meet the single guiding mode condition as revealed above. After the cleaning of the structured waveguide surfaces, the samples were covered with a layer of SiO_2 of a thickness $1.5 \mu\text{m}$. The samples were then diced into a dimension of 5 mm in length and 1.5 mm in width for a better handling, though the longest length of the couplers (l_{ac} , the length of the three-waveguide section) in the samples is only 2 mm . Finally, the end facets (*y* surfaces) of the samples were polished for conducting the optical tests; the fabrication of the LNOI waveguide adiabatic couplers were accomplished. More detailed processing steps and information for the LNOI waveguide structuring with the IBE method can be found in [28]. Figure 3(a) shows the SEM image of a portion of the ridge waveguides (with $w = 0.6 \mu\text{m}$ and $h = 0.6 \mu\text{m}$) structured in a LNOI test sample (before the overcoating of the SiO_2 layer),

revealing a typical sidewall condition of our fabricated waveguides. An image for revealing the three-waveguide configuration of an accomplished sample has been taken with the laser scanning confocal microscopy (a non-destructive imaging method), as shown in Fig. 3(b). Since the accomplished sample has been over-coated with a SiO₂ cladding layer (1.5- μ m thickness), the width of the strip structure shown in the image (Fig. 3(b)) will be broader than the actual width ($w = 1\ \mu$ m) of the thin-film LN waveguides underneath.

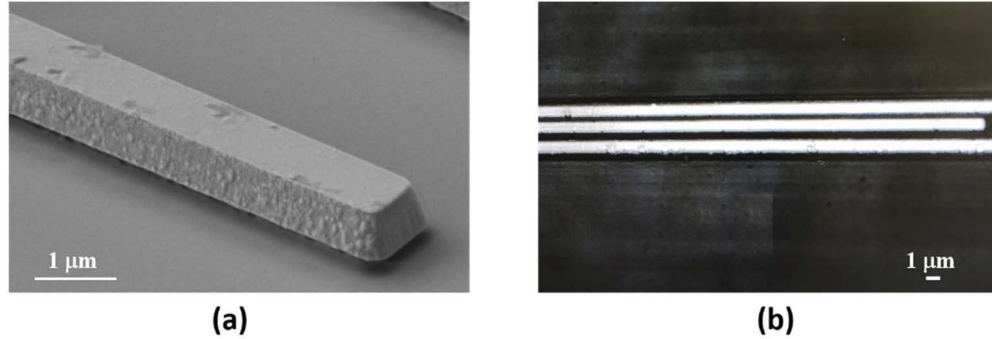


Fig. 3. (a) SEM image of a portion of the ridge waveguides (with $w = 0.6\ \mu\text{m}$ and $h = 0.6\ \mu\text{m}$) structured in a LNOI test sample. The images were taken before the overcoating of the SiO₂ layer. (b) Laser scanning confocal microscopic image of a portion of an accomplished sample (with $w = 1\ \mu\text{m}$ and $h = 0.6\ \mu\text{m}$) used in this study. Note the structure has been over-coated with a 1.5- μm thick SiO₂ cladding layer.

The fabricated LNOI adiabatic coupler samples were then characterized in an optical testbed using a continuous wave linearly polarized external cavity laser having a wavelength tuning range from 1450 to 1650 nm as the light source. The source, followed by an erbium doped fiber amplifier (EDFA) and a polarization controller, is then butt coupled into the LNOI sample in a counterintuitive coupling scheme (i.e., coupled to $\mathbf{Wg}_1$ as schematically shown in Fig. 1(b)) through a tapered and lensed fiber with a focal spot size of 2.8 μm and a working distance of 15 μm . The LNOI chip is mounted on a high-precision electric-controlled multi-axis translational stage. The light output from the LNOI chip was collected by a 40 $\times$ objective lens and then characterized using a IR CCD-based mode profiler and a power meter. From the measurements, we found among our fabricated LNOI devices, the waveguides with $w = 1\ \mu\text{m}$ and $h = 0.6\ \mu\text{m}$ reach the lowest insertion losses (of, e.g., 12.2 and 13.3 dB for TE- and TM-like modes, respectively, in a 2-mm long waveguide). Figures 4(a) and 4(b) show the measured output mode intensity profiles of a 2-mm long LNOI adiabatic coupler with $w = 1\ \mu\text{m}$, $h = 0.6\ \mu\text{m}$, $\Delta d = 1\ \mu\text{m}$, and $d_{23}(-l_{ac}/2) = 1\ \mu\text{m}$ (a representative sample with longer coupler length) when TE-like modes at 1500 and 1550 nm, respectively, are initially excited in $\mathbf{Wg}_1$. It clearly shows only one mode (exited from $\mathbf{Wg}_3$) is well sensed by the CCD, implying the majority of the input power in a counterintuitive coupling scheme has transferred from $\mathbf{Wg}_1$ to $\mathbf{Wg}_3$ without obvious excitation of $\mathbf{Wg}_2$. For comparison, the output modes of a shorter coupler of $l_{ac} = 0.4\ \text{mm}$ with $\Delta d = 0.3\ \mu\text{m}$, and $d_{23}(-l_{ac}/2) = 1.7\ \mu\text{m}$ (a representative sample for a shorter coupler) are also measured when excited by a 1525 nm TE polarized wave, as shown in Fig. 4(c). It can be observed that two modes appear with one output from $\mathbf{Wg}_3$ being much brighter than that from $\mathbf{Wg}_1$. The observation of an incomplete power transfer with a shorter coupler can be attributed to its smaller adiabaticity as discussed earlier and agrees with that predicted by the simulation shown in Fig. 2 (ith respect to the high- η (red color) regions of the distributions shown in Figs. 2(b) and 2(c)). Because of the gain limit of the employed CCD (Spiricon, SP 90419), we failed to measure the information of TM-like mode profiles because of the higher loss of this polarization mode.

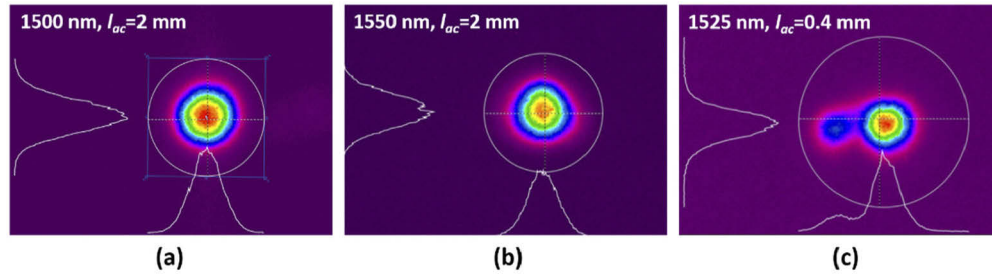


Fig. 4. Measured output mode intensity profiles of the 2-mm long LNOI adiabatic coupler at (a) 1500 nm and (b) 1550 nm and of (c) the 0.4-mm long coupler at 1525 nm. All shown are TE-like modes.

The loss of the optical coupling between the tapered lensed fiber and the LNOI coupler is estimated to be ~ 3.1 dB per facet, calculated based on the mode matching between the measured (TE-like) waveguide mode and the derived output beam mode of the tapered lensed fiber at the known working distance (between the fiber and the chip) [30]. With the measured insertion losses (12.2 dB for the TE-like mode), we estimate our device has a propagation loss of ~ 1.2 dB/mm. The estimated much higher propagation loss (especially when compared to the state-of-the-art values [21]) can be attributable to the underestimated coupling loss (when referring to a measured coupling loss (>5 dB per facet) reported in several recent studies [19,20]) as our estimation has assumed ideal conditions including that the device has perfectly polished optical facets and the optical alignment between the fiber and the nanophotonic chip is perfectly optimized. Further reduction of the coupling loss of our LNOI devices can be pursued by employing an advanced mode-matching scheme using such as an adiabatically tapered single mode fiber [31] or a linear inverse tapered LNOI waveguide [32].

Besides having a high light transfer efficiency, an adiabatic coupler features also broad bandwidth. The insets in Figs. 5(a) and 5(b) show the measured (solid lines) and calculated (solid dots) normalized output power from the exit ports of Wg_1 and Wg_3 of the 2-mm long LNOI adiabatic coupler as a function of wavelength in the experimental spectral range (1450–1650 nm) for TE-like and TM-like modes, respectively. In terms of the measurement data, a coupling efficiency of $>97\%$ (or >15 dB splitting ratio) over a broad coupling bandwidth (~ 200 and ~ 160 nm for TE-like and TM-like modes, respectively) across telecom S, C, and L bands is obtained. In terms of the fitted data, the device features even broader bandwidth at 15 dB, which are ~ 960 and ~ 800 nm for TE-like and TM-like modes, respectively, as the results calculated over a broader spectral range shown in Figs. 5(a) and 5(b). The measured and calculated normalized output power for a $l_{ac} = 0.4$ mm LNOI coupler is also shown (for the TE-like mode as a representative) in Fig. 5(c) for comparison. The coupling efficiency and bandwidth have been reduced to $\sim 90\%$ and ~ 80 nm, respectively, in the measurement range. The result manifests that a better adiabatic light transfer behavior can be expected from an adiabatic coupler with a longer length. The performance of these realized devices (such as the waveguide loss and coupling behavior) can be further enhanced if the waveguide structure quality (such as the sidewall roughness and angle) can be further improved. The influence of the waveguide sidewall angle on the performance of a nanophotonic element has been widely studied [33]. In general, in addition to the loss reduction, the improvement of the angled sidewalls from the structured waveguides will lead to a more controlled field coupling between (densely spaced) waveguides and contribute to the realization of a higher element-density circuits. The footprint of the realized LNOI adiabatic coupler ($\sim 3 \mu\text{m} \times 2 \text{ mm}$) has been reduced by $>99\%$ when compared to the bulk LN ones ($\sim 20 \mu\text{m} \times 50 \text{ mm}$) we have studied [15]. Further with the use of an appropriate

cross-sectional dimension of the thin-film ridge waveguides ($w = 1 \mu\text{m}$ and $h = 0.6 \mu\text{m}$), we achieve the most compact adiabatic coupler ever in LN platform working for both the TE and TM polarization modes over an ultra-broad bandwidth, underlining the high flexibility of performing dispersion engineering in LNOI waveguides. With diffused LN waveguides, two 5-cm long adiabatic couplers with different parameters designed for the two respective polarization modes would have to be used [15]. According to the data shown in Figs. 5(a) and 5(b), we plot the measured coupling efficiency (in decibel scale) versus wavelength for both polarization modes for the 2-mm long coupler, as shown in Fig. 5(d). The result shows the two curves for the coupling efficiency of the TE and TM polarization modes are well overlapped in a spectral range from ~ 1580 to ~ 1630 nm with a high splitting ratio of ~ 20 dB, indicating the LNOI adiabatic coupler can be a broadband non-polarizing mode coupler (NPMC) and further be a broadband non-polarizing (~ 3 dB) beam splitter when two of the NPMCs are combined into a symmetric five-waveguide adiabatic coupling system [34] (the structure presented in Ref. [34], however, supports only one polarization mode).

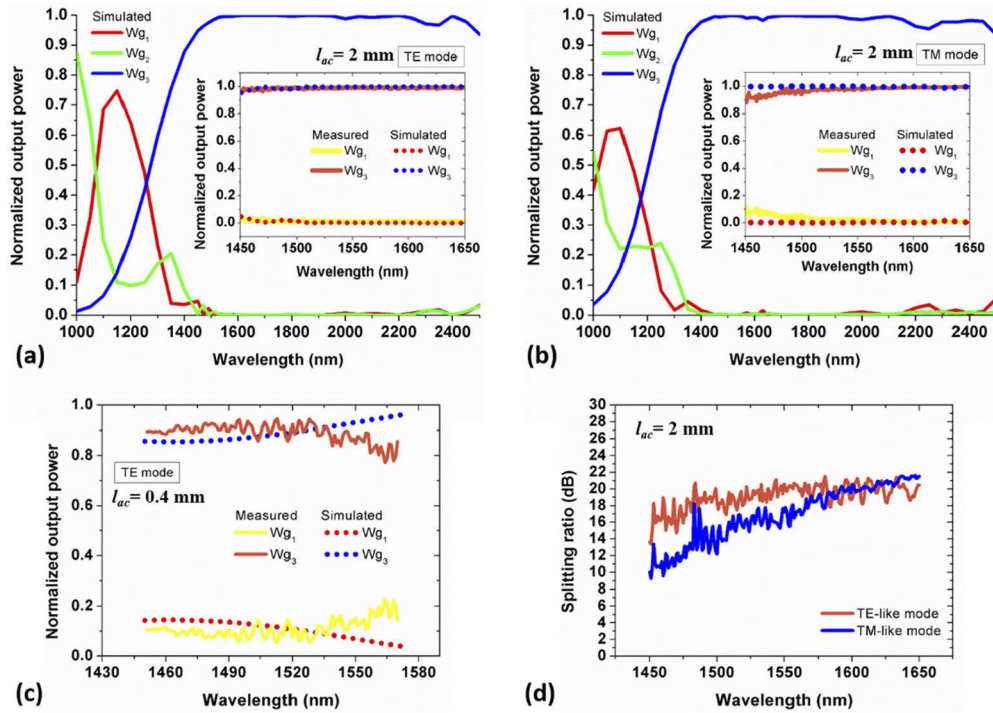


Fig. 5. Calculated normalized output power from the exit ports of Wg_1 , Wg_2 , and Wg_3 of the 2-mm long LNOI adiabatic coupler as a function of wavelength for (a) TE-like and (b) TM-like modes. The insets show the measured (solid lines) and calculated (solid dots) normalized output power from the exit ports of Wg_1 and Wg_3 of the 2-mm long LNOI adiabatic coupler in the experimental spectral range. (c) Measured (solid lines) and calculated (solid dots) normalized output power from the exit ports of Wg_1 and Wg_3 of the 0.4-mm long LNOI adiabatic coupler in the experimental spectral range. (d) Measured splitting ratios of the 2-mm long coupler versus wavelength for both polarization modes.

4. Conclusion

We have realized ultra-compact, ultra-broadband adiabatic directional couplers in the LNOI waveguide platform. A high-adiabaticity light passage process has been successfully observed in

a 2-mm long three-waveguide system, consisting of two parallel outer waveguides 3- μm apart and an inclined intermediate waveguide as the mediator. We measured from this device coupling bandwidths of ~ 200 and ~ 160 nm for TE-like and TM-like modes, respectively, in the laser tuning range in the telecom band at a >15 dB splitting ratio (or $>97\%$ coupling efficiency), which is in good agreement with the simulation results. The simulations predict an even broader bandwidth from 1490 to 2450 nm and from 1400 to 2200 nm for TE and TM polarization modes, respectively. Moreover, the footprint of the demonstrated LNOI adiabatic coupler has been miniaturized by $>99\%$ in contrast to the bulk LN counterparts of similar light transfer characteristics. This technology can be further applied to build many useful photonic elements such as a broadband non-polarizing beam splitter and is of great interest to many applications related to integrated photonic quantum circuits.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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